

Infinitesimal Higher-Order Spectral Variations in Rectangular Real Random Matrices

Róisín Luo*

Irish National Centre for Research Training in AI (CRT-AI)
University of Galway, Ireland

Abstract

We present a theoretical framework for deriving the general n -th order Fréchet derivatives of singular values in real rectangular matrices, by leveraging reduced resolvent operators from Kato’s analytic perturbation theory for self-adjoint operators. Deriving closed-form expressions for higher-order derivatives of singular values is notoriously challenging through standard matrix-analysis techniques. To overcome this, we treat a real rectangular matrix as a compact operator on a finite-dimensional Hilbert space, and embed the rectangular matrix into a block self-adjoint operator so that non-symmetric perturbations are captured. Applying Kato’s asymptotic eigenvalue expansion to this construction, we obtain a general, closed-form expression for the infinitesimal n -th order spectral variations. Specializing to $n = 2$ and deploying on a Kronecker-product representation with matrix convention yield the Hessian of a singular value, not found in literature. By bridging abstract operator-theoretic perturbation theory with matrices, our framework equips researchers with a practical toolkit for higher-order spectral sensitivity studies in random matrix applications (*e.g.* adversarial perturbation in deep learning).

1 Introduction

Singular values lie at the heart of modern matrix analysis, encapsulating key spectral information such as operator norms, conditioning, and rank structure, and underpinning applications across numerical linear algebra, data science, control theory, and mathematical physics [Horn and Johnson, 2012, Edelman and Rao, 2005]. In random matrix theory, they govern limiting distributions (*e.g.* Marchenko–Pastur laws [Marchenko and Pastur, 1967]), edge fluctuations (Tracy–Widom laws [Tracy and Widom, 1994]), and local eigenvalue spacings [Mehta, 2004]. In deep learning, the properties of matrix spectra govern robustness in both parameter spaces and input spaces [Szegedy et al., 2014, Goodfellow et al., 2014, Yin et al., 2019, Luo et al., 2024b,a]. Although the first-order derivatives (Jacobian) of singular values are well-known [Stewart and Sun, 1990, Wedin, 1972], explicit closed-form expressions for second- and higher-order derivatives are absent from the literature. Directly deriving higher-order derivatives of singular values via matrix analysis is challenging due to the intricate interplay among spectral gaps, left-singular and right-singular subspaces, and left-null and right-null subspaces.

Matrices are bounded linear operators on a finite-dimensional Hilbert space, which allows us to leverage Kato’s analytic perturbation theory [Kato, 1995] for self-adjoint operators. However a rectangular matrix $A \in \mathbb{R}^{m \times n}$ is not a self-adjoint linear operator since $A \neq A^\top$. Hence we can not directly apply Kato’s theory on rectangular matrices. To capture the perturbations from all subspaces

*Correspondence to: roisincrtai@gmail.com

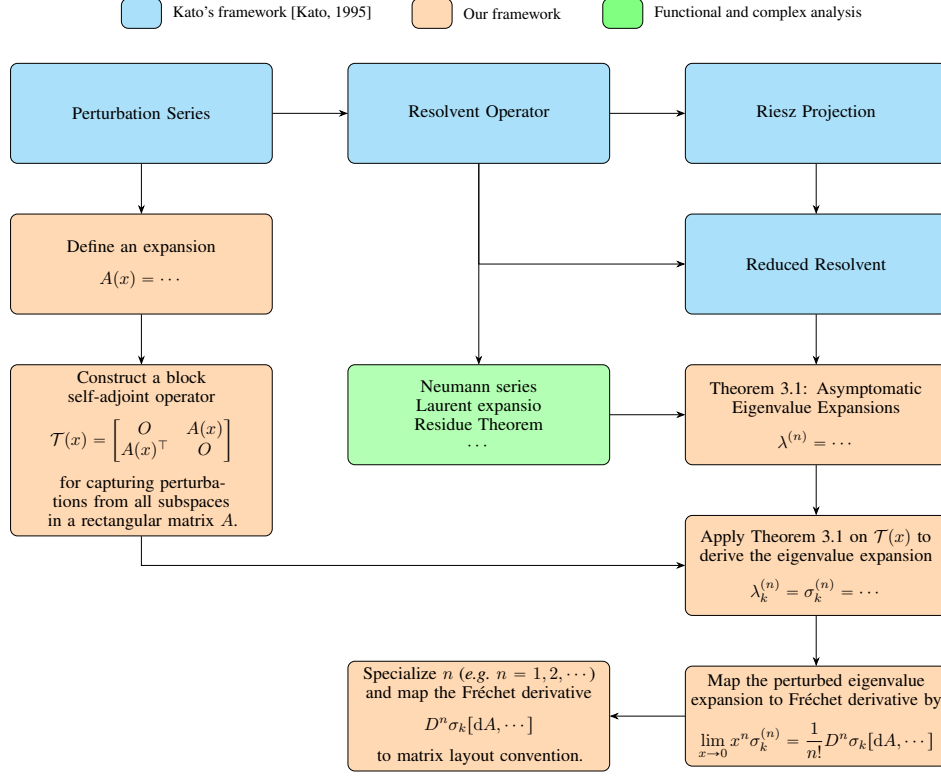


Figure 1: Theoretical framework for infinitesimal spectral variations in real rectangular matrices. We employ Kato’s analytic perturbation framework [Kato, 1995], which is originally formulated for perturbations of self-adjoint operators. To study the perturbations in rectangular matrices, we construct a block self-adjoint operator that captures perturbations in all subspaces (*i.e.* left-singular, right-singular, left-null, right-null subspaces). Applying Kato’s asymptotic eigenvalue expansions to this block operator, we then derive a n -th order eigenvalue expansion series, and relate with Fréchet derivatives. Finally, by specializing and mapping to matrix-layout conventions, we thus can generally derive arbitrary order derivatives of singular values for rectangular matrices.

(*i.e.* left-singular, right-singular, left-null and right-null subspaces), we construct a block operator:

$$\mathcal{T} := \begin{bmatrix} O & A \\ A^\top & O \end{bmatrix}, \quad (1)$$

which is self-adjoint since:

$$\mathcal{T}^\top := \begin{bmatrix} O & A \\ A^\top & O \end{bmatrix}^\top = \begin{bmatrix} O^\top & (A^\top)^\top \\ A^\top & O^\top \end{bmatrix} = \begin{bmatrix} O & A \\ A^\top & O \end{bmatrix} = \mathcal{T}. \quad (2)$$

We then first derive the asymptotic eigenvalue expansion of $\mathcal{T}$ up to n -th order under holomorphic perturbations (Theorem 3.1). By relating the n -th order expansion with its n -th order Fréchet derivative, we can obtain the n -th order Fréchet derivative. Finally, we deploy the n -th order Fréchet derivative with matrix or tensor layout conventions. A scheme of our framework is depicted in Figure 1. In particular, the first order case ($n = 1$) recovers the well-known singular-value Jacobian; while the second-order case ($n = 2$) produces the singular-value Hessian with a Kronecker-product representation, which is not previously available. By bridging the abstract operator-theoretic expansions with matrices, our framework provides a toolkit for arbitrary-order singular-value sensitivity analysis.

Contributions. This paper makes the following contributions:

1. **Spectral Variations in Rectangular Matrices.** We present a unified operator-theoretic framework for analyzing n -th order spectral variations in real rectangular matrices (see

Figure 1). This approach enables systematic higher-order sensitivity analysis in random matrix settings and extends classical perturbation theory to a broader asymmetric context.

2. **Singular-Value Hessian.** Specializing $n = 2$ of our general result obtains second-order derivative (Hessian) of singular value with a Kronecker-product representation, not found in the literature.

2 Fréchet Derivative and Layout Convention

To deploy the results from the abstract operator theory into matrices, we introduce layout conventions such as matrices, derivative representations, etc. Let

$$A = \begin{pmatrix} A_{1,1} & A_{1,2} & \cdots & A_{1,n} \\ A_{2,1} & A_{2,2} & \cdots & A_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m,1} & A_{m,2} & \cdots & A_{m,n} \end{pmatrix} \in \mathbb{R}^{m \times n} \quad (3)$$

be a real random matrix with rank $r = \text{rank}(A)$. The A admits a singular value decomposition (SVD) [Horn and Johnson, 2012, Stewart and Sun, 1990] given by:

$$A = \sum_{k=1}^r \sigma_k u_k v_k^\top, \quad (4)$$

where $r = \text{rank}(A)$, u_k and v_k are the left and right singular vectors associated with singular value $\sigma_k \geq 0$.

Assumption 2.1 (Simplicity Assumption of Singular Values). We assume that the singular values of A are *simple*

$$\sigma_i \neq \sigma_j \quad \text{for all } i \neq j. \quad (5)$$

Remark 2.2. This *simple* assumption is essential for $u_i, v_j, \sigma_k \in C^\infty$ (i.e. maps are infinitely continuously differentiable). Under this assumption, singular values and singular vectors vary continuously differentiable with respect to the entries of A .

2.1 Matrix Fréchet Derivative as Multilinear Operator

Definition 2.3 (α -times continuously Fréchet derivative). Let

$$F : \mathbb{R}^{m \times n} \mapsto \mathbb{R}^{k \times l} \quad (6)$$

be α -times continuously Fréchet-differentiable (i.e. $F \in C^\alpha$). Its α -th derivative near $A \in \mathbb{R}^{m \times n}$ can be piece-wisely defined as:

$$\left[\nabla^\alpha F(A) \right]_{i,j; p_1 q_1 \cdots p_\alpha q_\alpha} = \frac{\partial^\alpha F_{i,j}(A)}{\partial A_{p_1 q_1} \cdots \partial A_{p_\alpha q_\alpha}}. \quad (7)$$

Then for any directions $H_1, \dots, H_\alpha \in \mathbb{R}^{m \times n}$, the multilinear action is given by the natural Frobenius-tensor inner-product [Kolda and Bader, 2009, Magnus and Neudecker, 2019]

$$D^\alpha F(A)[H_1, \dots, H_\alpha] = \langle \nabla^\alpha F(A), H_1 \otimes \cdots \otimes H_\alpha \rangle \quad (8)$$

$$= \left[\nabla^\alpha F(A) \right]_{i,j; p_1 q_1 \cdots p_\alpha q_\alpha} (H_1)_{p_1 q_1} \cdots (H_\alpha)_{p_\alpha q_\alpha}. \quad (9)$$

The $F(A + H)$ can be expanded by **multivariate Taylor expansion**

$$F(A + H) = \sum_{\beta=0}^{\alpha} \frac{1}{\beta!} D^\beta F(A)[H, \dots, H] + o(\|H\|^\alpha). \quad (10)$$

Theorem 2.4 (Uniqueness of the α -times Fréchet derivative [Rudin, 2021, Spivak, 2018]). *Suppose $F \in C^\alpha$ is differentiable up to order α . Then $D^\alpha F$ exists, is a symmetric α -linear map, and is unique. That is, there is no other α -linear operator satisfying the defining Taylor-remainder condition.*

2.1.1 Matrix-to-Scalar Functionals

Let

$$f : \mathbb{R}^{m \times n} \mapsto \mathbb{R} \quad (11)$$

be a functional mapping a real random matrix to a scalar. Suppose there exists:

$$df = D^1 f(A)[dA] = \langle \nabla f(A), dA \rangle = \left(\frac{\partial f}{\partial A} \right)^\top dA = \text{tr} \left[\left(\frac{\partial f}{\partial A} \right)^\top dA \right], \quad (12)$$

where $\frac{\partial f}{\partial A}$ and infinitesimal variation $dA \in \mathbb{R}^{m \times n}$ are piece-wisely defined as:

$$\frac{\partial f}{\partial A} = \begin{pmatrix} \frac{\partial f}{\partial A_{1,1}} & \frac{\partial f}{\partial A_{1,2}} & \cdots & \frac{\partial f}{\partial A_{1,n}} \\ \frac{\partial f}{\partial A_{2,1}} & \frac{\partial f}{\partial A_{2,2}} & \cdots & \frac{\partial f}{\partial A_{2,n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial A_{m,1}} & \frac{\partial f}{\partial A_{m,2}} & \cdots & \frac{\partial f}{\partial A_{m,n}} \end{pmatrix} \quad (13)$$

with **denominator layout convention**, and:

$$dA = \begin{pmatrix} dA_{1,1} & dA_{1,2} & \cdots & dA_{1,n} \\ dA_{2,1} & dA_{2,2} & \cdots & dA_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ dA_{m,1} & dA_{m,2} & \cdots & dA_{m,n} \end{pmatrix}. \quad (14)$$

2.1.2 Vectorized Matrix-to-Matrix Functions

Let

$$\text{vec} : \mathbb{R}^{m \times n} \mapsto \mathbb{R}^{mn} \quad (15)$$

be the vectorization with the **column-major convention**. Let

$$F : \mathbb{R}^{m \times n} \mapsto \mathbb{R}^{k \times l} \quad (16)$$

be a matrix-to-matrix map. Suppose there exists

$$\frac{\partial \text{vec}(F)}{\partial \text{vec}(A)} \in \mathbb{R}^{p \times q} \quad p = mn \quad \text{and} \quad q = kl, \quad (17)$$

piece-wisely defined as

$$\frac{\partial \text{vec}(F)}{\partial \text{vec}(A)} = \begin{pmatrix} \frac{\partial \text{vec}(F)_1}{\partial \text{vec}(A)_1} & \frac{\partial \text{vec}(F)_2}{\partial \text{vec}(A)_1} & \cdots & \frac{\partial \text{vec}(F)_q}{\partial \text{vec}(A)_1} \\ \frac{\partial \text{vec}(F)_1}{\partial \text{vec}(A)_2} & \frac{\partial \text{vec}(F)_2}{\partial \text{vec}(A)_2} & \cdots & \frac{\partial \text{vec}(F)_q}{\partial \text{vec}(A)_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \text{vec}(F)_1}{\partial \text{vec}(A)_p} & \frac{\partial \text{vec}(F)_2}{\partial \text{vec}(A)_p} & \cdots & \frac{\partial \text{vec}(F)_q}{\partial \text{vec}(A)_p} \end{pmatrix} \quad (18)$$

by using **denominator layout convention** on $\text{vec}(F)$ and $\text{vec}(A)$.

3 Asymptotic Eigenvalue Expansion

Theorem 3.1 (Asymptomatic Eigenvalue Expansion of Self-Adjoint Operator). *Let*

$$\mathcal{L}(X) = \{ T : X \mapsto X \mid X \text{ is a Banach space and } T \text{ is a bounded linear operator} \} \quad (19)$$

be the Banach space of bounded linear operators. Let

$$\mathcal{T}(x) = \mathcal{T}^{(0)} + \sum_{j=1}^{\infty} x^j \mathcal{T}^{(j)} \in \mathcal{L}(X) \quad (20)$$

be a holomorphic family of bounded operators. Suppose σ_0 is a simple singular value of $\mathcal{T}^{(0)}$, with normalized singular vector w ($\|w\| = 1$). Define the reduced resolvent

$$S = (\mathcal{T}^{(0)} - \sigma_0 I)^{-1} (I - ww^*). \quad (21)$$

Then there is a unique holomorphic branch $\sigma(x)$ of singular values of $\mathcal{T}(x)$ with $\sigma(0) = \sigma_0$. It admits the power series

$$\sigma(x) = \sum_{n=0}^{\infty} x^n \sigma^{(n)}, \quad \sigma^{(0)} = \sigma_0, \quad (22)$$

and for each integer $n \geq 1$,

$$\sigma^{(n)} = \sum_{p=1}^n (-1)^{p-1} \sum_{\substack{i_1 + \dots + i_p = n \\ i_j \geq 1}} \langle w, \mathcal{T}^{(i_1)} S \mathcal{T}^{(i_2)} S \dots S \mathcal{T}^{(i_p)} w \rangle. \quad (23)$$

Proof 3.2. Since σ_0 is a simple singular value of $\mathcal{T}^{(0)}$, analytic-perturbation theory ensures there is a unique singular-value branch $\sigma(x)$ and rank-one Riesz projector

$$P(x) = \frac{1}{2\pi i} \oint_{\Gamma} (\mathcal{T}(x) - zI)^{-1} dz, \quad (24)$$

where Γ is a small contour enclosing only σ_0 . In particular,

$$\sigma(x) P(x) = \frac{1}{2\pi i} \oint_{\Gamma} z (\mathcal{T}(x) - zI)^{-1} dz, \quad (25)$$

and if w is the normalized singular vector of $\mathcal{T}^{(0)}$, then

$$\sigma(x) - \sigma_0 = \langle w, \sigma(x) P(x) w \rangle - \sigma_0 = \frac{1}{2\pi i} \oint_{\Gamma} (z - \sigma_0) \langle w, (\mathcal{T}(x) - zI)^{-1} w \rangle dz. \quad (26)$$

Set $R_0(z) = (\mathcal{T}^{(0)} - zI)^{-1}$, then

$$\mathcal{T}(x) - zI = (\mathcal{T}^{(0)} - zI) \left(I + R_0(z) \sum_{j=1}^{\infty} x^j \mathcal{T}^{(j)} \right), \quad (27)$$

the Neumann series gives

$$(\mathcal{T}(x) - zI)^{-1} = R_0(z) \sum_{k=0}^{\infty} (-1)^k \left(R_0(z) \sum_{j=1}^{\infty} x^j \mathcal{T}^{(j)} \right)^k. \quad (28)$$

Hence

$$(\mathcal{T}(x) - zI)^{-1} = \sum_{k=0}^{\infty} (-1)^k \sum_{i_1, \dots, i_k \geq 1} x^{i_1 + \dots + i_k} R_0(z) \mathcal{T}^{(i_1)} R_0(z) \dots R_0(z) \mathcal{T}^{(i_k)} R_0(z). \quad (29)$$

Substituting into the contour integral for $\sigma(x) - \sigma_0$, only multi-indices satisfying $i_1 + \dots + i_k = n$ contribute to the coefficient of x^n . Therefore,

$$\sigma^{(n)} = \sum_{k=0}^{\infty} (-1)^k \sum_{\substack{i_1 + \dots + i_k = n \\ i_j \geq 1}} \frac{1}{2\pi i} \oint_{\Gamma} (z - \sigma_0) \langle w, R_0(z) \mathcal{T}^{(i_1)} R_0(z) \dots R_0(z) \mathcal{T}^{(i_k)} R_0(z) w \rangle dz. \quad (30)$$

Since $k \leq n$, the term with $k = 0$ vanishes for $n \geq 1$. Re-index $k = p - 1$, so $1 \leq p \leq n$:

$$\sigma^{(n)} = \sum_{p=1}^n (-1)^{p-1} \sum_{\substack{i_1 + \dots + i_p = n \\ i_j \geq 1}} \frac{1}{2\pi i} \oint_{\Gamma} (z - \sigma_0) \langle w, R_0(z) \mathcal{T}^{(i_1)} R_0(z) \dots R_0(z) \mathcal{T}^{(i_p)} R_0(z) w \rangle dz. \quad (31)$$

Use residue theorem:

$$\sigma^{(n)} = \sum_{p=1}^n (-1)^{p-1} \sum_{\substack{i_1+\dots+i_p=n \\ i_j \geq 1}} \text{Res}_{z=\sigma_0} \langle w, (z - \sigma_0) R_0(z) \mathcal{T}^{(i_1)} \dots R_0(z) w \rangle. \quad (32)$$

Near $z = \sigma_0$, the Laurent expansion of $R_0(z)$ is

$$R_0(z) = \frac{P}{z - \sigma_0} + S + \mathcal{O}(z - \sigma_0), \quad (33)$$

where $P = ww^*$. In the product

$$(z - \sigma_0) R_0 \mathcal{T}^{(i_1)} R_0 \dots R_0 \mathcal{T}^{(i_p)} R_0, \quad (34)$$

only two copies of R_0 must contribute $\frac{P}{z - \sigma_0}$, and the remaining $p - 1$ factors contribute S . Because $PS = 0$, the only nonzero Laurent coefficient at $(z - \sigma_0)^{-1}$ is

$$\frac{P \mathcal{T}^{(i_1)} S \mathcal{T}^{(i_2)} S \dots S \mathcal{T}^{(i_p)} P}{z - \sigma_0}. \quad (35)$$

Hence:

$$\text{Res}_{z=\sigma_0} \langle w, (z - \sigma_0) R_0(z) \mathcal{T}^{(i_1)} \dots R_0(z) w \rangle = \langle w, \mathcal{T}^{(i_1)} S \mathcal{T}^{(i_2)} S \dots S \mathcal{T}^{(i_p)} w \rangle \quad (36)$$

$$\implies \sigma^{(n)} = \sum_{p=1}^n (-1)^{p-1} \sum_{\substack{i_1+\dots+i_p=n \\ i_j \geq 1}} \langle w, \mathcal{T}^{(i_1)} S \mathcal{T}^{(i_2)} S \dots S \mathcal{T}^{(i_p)} w \rangle. \quad (37)$$

4 Infinitesimal Spectral Variations

We treat real matrices as linear, compact operators on Hilbert spaces. Let

$$A \in \mathbb{R}^{m \times n} \quad (38)$$

be a real random matrix. Let

$$A(x) = \sum_{k=0}^{\infty} x^k A^{(k)} \in \mathbb{R}^{m \times n} \quad (39)$$

be holomorphic perturbed operator near $x = 0$. The unperturbed matrix $A^{(0)}$ admits a SVD

$$A^{(0)} = U^{(0)} \Sigma^{(0)} (V^{(0)})^T, \quad (40)$$

where

$$\Sigma^{(0)} = \text{diag}(\sigma_1^{(0)}, \dots, \sigma_r^{(0)}, 0, \dots, 0), \quad \sigma_k^{(0)} > 0 \quad (41)$$

is a simple singular value, and

$$U^{(0)} \in \mathbb{R}^{m \times m}, \quad V^{(0)} \in \mathbb{R}^{n \times n} \quad (42)$$

are orthogonal matrices. The singular vectors satisfy:

$$A^{(0)} v_k^{(0)} = \sigma_k^{(0)} u_k^{(0)}, \quad (A^{(0)})^T u_k^{(0)} = \sigma_k^{(0)} v_k^{(0)}, \quad (43)$$

with $\|u_k^{(0)}\| = \|v_k^{(0)}\| = 1$.

A key challenge is that A is not a self-adjoint operator since:

$$A^\top \neq A, \quad (44)$$

so that we are not able to leverage the theoretical framework from [Kato, 1995] directly. To account for perturbations arising in all subspaces of A (*i.e.* left-singular, right-singular, left-null and right-null subspaces), we construct a block self-adjoint operator:

$$\mathcal{T}(x) = \begin{bmatrix} 0 & A(x) \\ A(x)^\top & 0 \end{bmatrix}, \quad (45)$$

satisfying

$$\mathcal{T}(x)^\top = \mathcal{T}(x). \quad (46)$$

Let

$$\mathcal{T}(x) = \sum_{j=0}^{\infty} x^j \mathcal{T}^{(j)} \quad (47)$$

be a perturbation expansion series at x near zero, with the unperturbed operator:

$$\mathcal{T}^{(0)} = \begin{bmatrix} 0 & A^{(0)} \\ (A^{(0)})^\top & 0 \end{bmatrix}, \quad (48)$$

and the perturbations:

$$\mathcal{T}^{(j)} = \begin{bmatrix} 0 & A^{(j)} \\ (A^{(j)})^\top & 0 \end{bmatrix}, \quad j \geq 1. \quad (49)$$

The eigenvalues of $\mathcal{T}^{(0)}$ are

$$\pm \sigma_i, \quad \text{for } i = 1, \dots, r, \quad (50)$$

associated with eigenvectors:

$$w_i^{(+)} = \frac{1}{\sqrt{2}} \begin{bmatrix} u_i \\ v_i \end{bmatrix}, \quad w_i^{(-)} = \frac{1}{\sqrt{2}} \begin{bmatrix} u_i \\ -v_i \end{bmatrix}, \quad (51)$$

and zero eigenvectors,

$$a_j = \begin{bmatrix} u_j \\ 0 \end{bmatrix}, \quad (\text{for } j = r+1, \dots, m), \quad (52)$$

and

$$b_j = \begin{bmatrix} 0 \\ v_j \end{bmatrix}, \quad (\text{for } j = r+1, \dots, n). \quad (53)$$

Remark 4.1. The normalization $\frac{1}{\sqrt{2}}$ for $w_i^{(+)}$ and $w_i^{(-)}$ ensures:

$$\|w_k^{(0)}\|_2^2 = \frac{1}{2} (\|u_k^{(0)}\|_2^2 + \|v_k^{(0)}\|_2^2) = \frac{1}{2} (1 + 1) = 1. \quad (54)$$

Since $w_k^{(+0)}$ and $w_k^{(-0)}$ are eigenvectors of $\mathcal{T}^{(0)}$, hence the identities hold:

$$\mathcal{T}^{(0)} w_k^{(+0)} = \sigma_k^{(0)} w_k^{(+0)}, \quad (55)$$

and

$$\mathcal{T}^{(0)} w_k^{(-0)} = -\sigma_k^{(0)} w_k^{(-0)}. \quad (56)$$

By definition, the spectral expansion of the reduced resolvent operator associated with the eigenvalue σ_k of $\mathcal{T}(x)$ is:

$$S_k = (\mathcal{T} - \sigma_k I)^{-1} (I - P_k), \quad P_k = w_k^{(+)} (w_k^{(+)})^\top. \quad (57)$$

with:

$$S_k = \sum_{i \neq k} \frac{w_i^{(+)} (w_i^{(+)})^\top}{\sigma_i - \sigma_k} + \sum_{i=1}^r \frac{w_i^{(-)} (w_i^{(-)})^\top}{-\sigma_i - \sigma_k} - \sum_{j=r+1}^m \frac{a_j a_j^\top}{\sigma_k^{(0)}} - \sum_{j=r+1}^n \frac{b_j b_j^\top}{\sigma_k^{(0)}}. \quad (58)$$

According to Theorem 3.1, we know the asymptotic expansion is,

$$\sigma_k(x) = \sum_{n=0}^{\infty} x^n \sigma_k^{(n)} \quad (59)$$

where for $n \geq 1$:

$$\sigma_k^{(n)} = \sum_{p=1}^n (-1)^{p-1} \sum_{\substack{i_1+\dots+i_p=n \\ i_j \geq 1}} \langle w_k, \mathcal{T}^{(i_1)} S \mathcal{T}^{(i_2)} \dots S \mathcal{T}^{(i_p)} w_k \rangle. \quad (60)$$

By Definition 2.3, we know

$$\sigma_k^{(n)} = \frac{1}{n!} \frac{d^n \sigma_k}{dx^n} \Big|_{x=0} = \frac{1}{n!} D^n \sigma_k \quad (61)$$

$$\implies \lim_{x \rightarrow 0} x^n \sigma_k^{(n)} [A^{(1)}, \dots, A^{(1)}] = \frac{1}{n!} \lim_{x \rightarrow 0} x^n D^n \sigma_k [A^{(1)}, \dots, A^{(1)}] \quad (62)$$

$$\implies \lim_{x \rightarrow 0} x^n \sigma_k^{(n)} [xA^{(1)}, \dots, xA^{(1)}] = \frac{1}{n!} \lim_{x \rightarrow 0} D^n \sigma_k [xA^{(1)}, \dots, xA^{(1)}] \quad (63)$$

$$\implies \sigma_k^{(n)} [dA, \dots, dA] = \frac{1}{n!} D^n \sigma_k [dA, \dots, dA] \quad (64)$$

$$\implies D^n \sigma_k [dA, \dots, dA] = n! \sigma_k^{(n)} [dA, \dots, dA] \quad (65)$$

where

$$dA := \lim_{x \rightarrow 0} x A^{(1)}. \quad (66)$$

Theorem 4.2 (The n -th Order Infinitesimal Spectral Variation). *For each integer $n \geq 1$,*

$$\sigma_k^{(n)} = \sum_{p=1}^n (-1)^{p-1} \sum_{\substack{i_1+\dots+i_p=n \\ i_j \geq 1}} \langle w_k, \mathcal{T}^{(i_1)} S_k \mathcal{T}^{(i_2)} \dots S_k \mathcal{T}^{(i_p)} w_k \rangle. \quad (67)$$

The Fréchet derivative of the singular value is unique and given by:

$$D^n \sigma_k [dA, \dots, dA] = n! \sigma_k^{(n)} [dA, \dots, dA] \quad (68)$$

where

$$dA := \lim_{x \rightarrow 0} x A^{(1)}. \quad (69)$$

5 Special Case ($n = 1$): Jacobian of Singular Value

Lemma 5.1 (Jacobian of singular value). *The Jacobian of a singular value is well-known in the literature [Stewart and Sun, 1990, Magnus and Neudecker, 2019] in the form*

$$\frac{\partial \sigma_k}{\partial A} = u_k v_k^\top = v_k \otimes u_k. \quad (70)$$

Proof. We now recover this result by applying Theorem 4.2 and specializing $n = 1$.

Consider the perturbation on A as:

$$A(x) = A + x A^{(1)}, \quad A^{(0)} = A, \quad dA = \lim_{x \rightarrow 0} x A^{(1)}. \quad (71)$$

Set $n = 1$ in Theorem 4.2:

$$\sigma_k^{(1)} = \langle w_k, \mathcal{T}^{(1)} w_k \rangle. \quad (72)$$

Consider:

$$\mathcal{T}^{(1)} w_k = \begin{bmatrix} 0 & A^{(1)} \\ (A^{(1)})^\top & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} u_k \\ v_k \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} \quad (73)$$

$$\implies \langle w_k, \mathcal{T}^{(1)} w_k \rangle = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} u_k \\ v_k \end{bmatrix} \right)^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} \quad (74)$$

$$\implies \langle w_k, \mathcal{T}^{(1)} w_k \rangle = \frac{1}{2} \left[u_k^\top A^{(1)} v_k + v_k^\top (A^{(1)})^\top u_k \right] \quad (75)$$

$$\implies \langle w_k, \mathcal{T}^{(1)} w_k \rangle = u_k^\top A^{(1)} v_k \quad (76)$$

$$\implies \sigma_k^{(1)} = u_k^\top A^{(1)} v_k. \quad (77)$$

Therefore,

$$D\sigma_k[dA] = \sigma_k^{(1)}[dA] = \lim_{x \rightarrow 0} x\sigma_k^{(1)} \quad (78)$$

$$= \lim_{x \rightarrow 0} u_k^T x A^{(1)} v_k \quad (79)$$

$$= u_k^\top dA v_k \quad (80)$$

$$= \text{tr}(u_k^\top dA v_k) \quad (81)$$

$$= \text{tr}(v_k u_k^\top dA) \quad (82)$$

$$= \text{tr} \left(\left(\frac{\partial \sigma_k}{\partial A} \right)^T dA \right). \quad (83)$$

Use the identities:

1. $\text{vec}(BVA^\top) = (A \otimes B) \text{vec}(V)$,
2. $A^\top B = \text{tr}(A^\top B)$,
3. $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$,

thus:

$$\frac{\partial \sigma_k}{\partial A} = u_k v_k^\top = v_k \otimes u_k. \quad (84)$$

□

6 Special Case ($n = 2$): Hessian of Singular Value

We now derive the Hessian of the singular value (Lemma 6.1), with the layout

$$D^2[dA, dA] = \text{vec}(dA)^\top \frac{\partial}{\partial \text{vec}(A)} \text{vec} \left(\frac{\partial \sigma_k}{\partial A} \right) \text{vec}(dA), \quad (85)$$

using the Theorem 4.2 by specializing $n = 2$. To the best of our knowledge, this result is not literature.

Lemma 6.1 (Hessian of singular value). *The Hessian of a singular value is given as:*

$$\frac{\partial}{\partial \text{vec}(A)} \text{vec} \left(\frac{\partial \sigma_k}{\partial \text{vec}(A)} \right) = \underbrace{\sum_{i \neq k, i \leq m} \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} (v_k \otimes u_i) (v_k \otimes u_i)^\top}_{\text{left}} + \quad (86)$$

$$\underbrace{\sum_{j \neq k, j \leq n} \frac{\sigma_k}{\sigma_k^2 - \sigma_j^2} (v_j \otimes u_k) (v_j \otimes u_k)^\top}_{\text{right}} + \quad (87)$$

$$\underbrace{\sum_{l \neq k, l \leq r} \frac{\sigma_l}{\sigma_k^2 - \sigma_l^2} \left[(v_k \otimes u_l) (v_l \otimes u_k)^\top + (v_l \otimes u_k) (v_k \otimes u_l)^\top \right]}_{\text{left-right interaction}} \quad (88)$$

with Kronecker-product representation.

Proof 6.2. Consider the perturbation on A as:

$$A(x) = A + xA^{(1)}, \quad A^{(0)} = A, \quad dA = \lim_{x \rightarrow 0} xA^{(1)}. \quad (89)$$

Set $n = 2$ in Theorem 4.2:

$$\sigma_k^{(2)} = \sum_{p=1}^2 (-1)^{p-1} \sum_{\substack{i_1 + \dots + i_p = 2 \\ i_j \geq 1}} \langle w_k, \mathcal{T}^{(i_1)} S \mathcal{T}^{(i_2)} \dots S \mathcal{T}^{(i_p)} w_k \rangle \quad (90)$$

$$= \sigma_k^{(2,p=1)} + \sigma_k^{(2,p=2)} \quad (91)$$

where

$$\sigma_k^{(2,p=1)} := \langle w_k, \mathcal{T}^{(2)} w_k \rangle, \quad (92)$$

and

$$\sigma_k^{(2,p=2)} := -\langle w_k, \mathcal{T}^{(1)} S_k \mathcal{T}^{(1)} w_k \rangle. \quad (93)$$

Part 6.2.1 (Compute the term $\sigma_k^{(2,p=1)}$).

$$\mathcal{T}^{(2)} w_k = \begin{bmatrix} 0 & A^{(2)} \\ (A^{(2)})^\top & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} u_k \\ v_k \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(2)} v_k \\ (A^{(2)})^\top u_k \end{bmatrix} \quad (94)$$

$$\implies \langle w_k, \mathcal{T}^{(2)} w_k \rangle = \frac{1}{2} \left[u_k^\top A^{(2)} v_k + (v_k)^\top (A^{(2)})^\top u_k \right] \quad (95)$$

$$\implies \langle w_k, \mathcal{T}^{(2)} w_k \rangle = u_k^\top A^{(2)} v_k \quad (96)$$

$$\xrightarrow{A^{(2)}=O} \sigma_k^{(2,p=1)} = u_k^\top A^{(2)} v_k = 0. \quad (97)$$

Part 6.2.2 (Compute the term $\sigma_k^{(2,p=2)}$). Consider the spectral expansion of the reduced resolvent:

$$S_k = \sum_{i \neq k} \frac{w_i^{(+)} (w_i^{(+)})^\top}{\sigma_i - \sigma_k} + \sum_{i=1}^r \frac{w_i^{(-)} (w_i^{(-)})^\top}{-\sigma_i - \sigma_k} - \sum_{j=r+1}^m \frac{a_j a_j^\top}{\sigma_k^{(0)}} - \sum_{j=r+1}^n \frac{b_j b_j^\top}{\sigma_k^{(0)}} \quad (98)$$

$$= \underbrace{\sum_{i \neq k} \frac{1}{2} \frac{\begin{bmatrix} u_k \\ v_k \end{bmatrix} \begin{bmatrix} u_k^\top & v_k^\top \end{bmatrix}}{\sigma_i - \sigma_k}}_{\text{positive eigenspaces}} + \underbrace{\sum_{i=1}^r \frac{1}{2} \frac{\begin{bmatrix} u_k \\ -v_k \end{bmatrix} \begin{bmatrix} u_k^\top & -v_k^\top \end{bmatrix}}{-\sigma_i - \sigma_k}}_{\text{negative eigenspaces}} - \underbrace{\sum_{j=r+1}^m \frac{\begin{bmatrix} u_k \\ 0 \end{bmatrix} \begin{bmatrix} u_k^\top & 0 \end{bmatrix}}{\sigma_k^{(0)}}}_{\text{left-null eigenspaces}} - \underbrace{\sum_{j=r+1}^n \frac{\begin{bmatrix} 0 \\ v_k \end{bmatrix} \begin{bmatrix} 0 & v_k^\top \end{bmatrix}}{\sigma_k^{(0)}}}_{\text{right-null eigenspaces}}. \quad (99)$$

Apply S_k on

$$S_k \mathcal{T}^{(1)} w_k = S_k \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix}, \quad (100)$$

and discuss the terms in terms of *positive eigenspaces*, *negative eigenspaces*, *left-null eigenspaces* and *right-null eigenspaces*:

1. Positive eigenspaces.

$$(w_i^{(+)})^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} u_i^\top & v_i^\top \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} \\ \implies (w_i^{(+)})^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{2} \left[u_i^\top A^{(1)} v_k + v_i^\top (A^{(1)})^\top u_k \right] \quad (101)$$

$$\xrightarrow{v_i^\top (A^{(1)})^\top u_k = u_k^\top A^{(1)} v_i} (w_i^{(+)})^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{2} \left[u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i \right] \quad (102)$$

$$\implies \frac{w_i^{(+)} (w_i^{(+)})^\top}{\sigma_i - \sigma_k} \cdot \mathcal{T}^{(1)} w_k = \frac{1}{\sqrt{2}} \cdot \frac{u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i}{2(\sigma_i - \sigma_k)} \begin{bmatrix} u_i \\ v_i \end{bmatrix}. \quad (103)$$

2. Negative eigenspaces.

$$(w_i^{(-)})^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} u_i^\top & -v_i^\top \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} \\ \implies (w_i^{(-)})^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{2} \left[u_i^\top A^{(1)} v_k - v_i^\top (A^{(1)})^\top u_k \right] \quad (104)$$

$$\xrightarrow{v_i^\top (A^{(1)})^\top u_k = u_k^\top A^{(1)} v_i} (w_i^{(-)})^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{2} \left[u_i^\top A^{(1)} v_k - u_k^\top A^{(1)} v_i \right] \quad (105)$$

$$\implies \frac{w_i^{(-)} (w_i^{(-)})^\top}{-\sigma_i - \sigma_k} \cdot \mathcal{T}^{(1)} w_k = \frac{1}{\sqrt{2}} \cdot \frac{u_i^\top A^{(1)} v_k - u_k^\top A^{(1)} v_i}{2(-\sigma_i - \sigma_k)} \begin{bmatrix} u_i \\ -v_i \end{bmatrix}. \quad (106)$$

3. Left-null eigenspaces.

$$a_j a_j^\top = \begin{bmatrix} u_j \\ 0 \end{bmatrix} \begin{bmatrix} u_j^\top & 0 \end{bmatrix} = \begin{bmatrix} u_j u_j^\top & 0 \\ 0 & 0 \end{bmatrix} \quad (107)$$

$$\implies a_j a_j^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} u_j u_j^\top A^{(1)} v_k \\ 0 \end{bmatrix} \quad (108)$$

$$\implies -\frac{a_j a_j^\top}{\sigma_k^{(0)}} \cdot \mathcal{T}^{(1)} w_k = -\frac{1}{\sqrt{2}} \cdot \frac{u_j^\top A^{(1)} v_k}{\sigma_k} \begin{bmatrix} u_j \\ 0 \end{bmatrix}. \quad (109)$$

4. Right-null eigenspaces.

$$b_j b_j^\top = \begin{bmatrix} 0 \\ v_j \end{bmatrix} \begin{bmatrix} 0 & v_j^\top \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & v_j v_j^\top \end{bmatrix} \quad (110)$$

$$\implies b_j b_j^\top \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ v_j v_j^\top A^{(1)} u_k \end{bmatrix} \quad (111)$$

$$\implies -\frac{b_j b_j^\top}{\sigma_k^{(0)}} \cdot \mathcal{T}^{(1)} w_k = -\frac{1}{\sqrt{2}} \cdot \frac{u_k^\top A^{(1)} v_j}{\sigma_k} \begin{bmatrix} 0 \\ v_j \end{bmatrix}. \quad (112)$$

Thus,

$$\sigma_k^{(2,p=2)} = -\langle w_k, \mathcal{T}^{(1)} S_k \mathcal{T}^{(1)} w_k \rangle \quad (113)$$

$$= \langle w_k, \mathcal{T}^{(1)} S_k \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} \rangle \quad (114)$$

$$= \langle w_k, \mathcal{T}^{(1)} S_k \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix} \rangle \quad (115)$$

$$= \left(\frac{1}{\sqrt{2}} \begin{bmatrix} u_k \\ v_k \end{bmatrix} \right)^\top \mathcal{T}^{(1)} S_k \frac{1}{\sqrt{2}} \begin{bmatrix} A^{(1)} v_k \\ (A^{(1)})^\top u_k \end{bmatrix}. \quad (116)$$

Part 6.2.3 (Contributions in $\langle w_k, \mathcal{T}^{(1)} S_k \mathcal{T}^{(1)} w_k \rangle$). We compute the contributions in $\mathcal{T}^{(1)} S_k$:

1. Contributions in positive eigenspaces.

$$* := \mathcal{T}^{(1)} \left(\frac{1}{\sqrt{2}} \cdot \frac{u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i^{(0)}}{2(\sigma_i - \sigma_k)} \begin{bmatrix} u_i \\ v_i \end{bmatrix} \right) \quad (117)$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i}{2(\sigma_i - \sigma_k)} \begin{bmatrix} A^{(1)} v_i \\ (A^{(1)})^\top u_i \end{bmatrix} \quad (118)$$

$$\implies \langle w_k, * \rangle = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} u_k \\ v_k \end{bmatrix} \right)^\top \frac{1}{\sqrt{2}} \cdot \frac{u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i}{2(\sigma_i - \sigma_k)} \begin{bmatrix} A^{(1)} v_i \\ (A^{(1)})^\top u_i \end{bmatrix} \quad (119)$$

$$\implies \langle w_k, * \rangle = \left(\frac{1}{\sqrt{2}} \right)^2 \cdot \frac{u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i}{2(\sigma_i - \sigma_k)} \cdot \left[u_k^\top A^{(1)} v_i + v_k^\top (A^{(1)})^\top u_i \right] \quad (120)$$

$$\implies \langle w_k, * \rangle = \frac{1}{4(\sigma_i - \sigma_k)} \left[u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i \right] \left[u_k^\top A^{(1)} v_i + u_i^\top A^{(1)} v_k \right] \quad (121)$$

$$\implies \langle w_k, * \rangle = \frac{1}{4(\sigma_i - \sigma_k)} \left[u_i^\top A^{(1)} v_k + u_k^\top A^{(1)} v_i \right]^2 \quad (122)$$

$$\implies \langle w_k, * \rangle = \frac{1}{4(\sigma_i - \sigma_k)} \left[\left[u_i^\top A^{(1)} v_k \right]^2 + 2u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \left[u_k^\top A^{(1)} v_i \right]^2 \right] \quad (123)$$

$$\xrightarrow{\text{sum over } i \neq k} \sum_{i=1}^r \frac{1}{4(\sigma_i - \sigma_k)} \left[\left[u_i^\top A^{(1)} v_k \right]^2 + 2u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \left[u_k^\top A^{(1)} v_i \right]^2 \right] \quad (124)$$

2. Contributions in negative eigenspaces.

$$* := \mathcal{T}^{(1)} \left(\frac{1}{\sqrt{2}} \cdot \frac{u_i^\top A^{(1)} v_k - u_k^\top A^{(1)} v_i}{2(-\sigma_i - \sigma_k)} \begin{bmatrix} u_i \\ -v_i \end{bmatrix} \right) \quad (125)$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{u_i^\top A^{(1)} v_k - u_k^\top A^{(1)} v_i}{2(-\sigma_i - \sigma_k)} \begin{bmatrix} A^{(1)}(-v_i) \\ (A^{(1)})^\top u_i \end{bmatrix} \quad (126)$$

$$\implies \langle w_k, * \rangle = \frac{1}{2} \cdot \frac{u_i^\top A^{(1)} v_k - u_k^\top A^{(1)} v_i}{2(-\sigma_i - \sigma_k)} \left[u_k^\top A^{(1)}(-v_i) + v_k^\top (A^{(1)})^\top u_i \right] \quad (127)$$

$$\implies \langle w_k, * \rangle = \frac{1}{4(-\sigma_i - \sigma_k)} \left[u_i^\top A^{(1)} v_k - u_k^\top A^{(1)} v_i \right] \left[-(u_k)^\top A^{(1)} v_i + u_i^\top A^{(1)} v_k \right] \quad (128)$$

$$\implies \langle w_k, * \rangle = \frac{1}{4(-\sigma_i - \sigma_k)} \left[u_i^\top A^{(1)} v_k - u_k^\top A^{(1)} v_i \right]^2 \quad (129)$$

$$\implies \langle w_k, * \rangle = \frac{1}{4(-\sigma_i - \sigma_k)} \left[\left[u_i^\top A^{(1)} v_k \right]^2 - 2u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \left[u_k^\top A^{(1)} v_i \right]^2 \right] \quad (130)$$

$$\xrightarrow{\text{sum over } i \neq k} \sum_{i=1}^r \frac{1}{4(-\sigma_i - \sigma_k)} \left[\left[u_i^\top A^{(1)} v_k \right]^2 - 2u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \left[u_k^\top A^{(1)} v_i \right]^2 \right]. \quad (131)$$

3. Contributions in left-null eigenspaces.

$$* := \mathcal{T}^{(1)} \left(-\frac{1}{\sqrt{2}} \cdot \frac{u_j^\top A^{(1)} v_k}{\sigma_k} \begin{bmatrix} u_j \\ 0 \end{bmatrix} \right) = -\frac{1}{\sqrt{2}} \cdot \frac{u_j^\top A^{(1)} v_k}{\sigma_k} \begin{bmatrix} 0 \\ (A^{(1)})^\top u_j \end{bmatrix} \quad (132)$$

$$\implies \langle w_k, * \rangle = -\frac{1}{2} \cdot \frac{u_j^\top A^{(1)} v_k}{\sigma_k} \left[u_k^\top \cdot 0 + v_k^\top (A^{(1)})^\top u_j \right] \quad (133)$$

$$\implies \langle w_k, * \rangle = -\frac{1}{2} \cdot \frac{u_j^\top A^{(1)} v_k u_j^\top A^{(1)} v_k}{\sigma_k} \quad (134)$$

$$\xrightarrow{\text{sum over } j=r+1, \dots, m} - \sum_{j=r+1}^m \frac{1}{2\sigma_k} \left[u_j^\top A^{(1)} v_k \right]^2. \quad (135)$$

4. Contributions in right-null eigenspaces.

$$* := \mathcal{T}^{(1)} \left(-\frac{1}{\sqrt{2}} \cdot \frac{u_k^\top A^{(1)} v_j}{\sigma_k} \begin{bmatrix} 0 \\ v_j \end{bmatrix} \right) = -\frac{1}{\sqrt{2}} \cdot \frac{u_k^\top A^{(1)} v_j}{\sigma_k} \begin{bmatrix} A^{(1)} v_j \\ 0 \end{bmatrix} \quad (136)$$

$$\implies \langle w_k, * \rangle = -\frac{1}{2} \cdot \frac{u_k^\top A^{(1)} v_j}{\sigma_k} \left[u_k^\top A^{(1)} v_j + v_k^\top \cdot 0 \right] \quad (137)$$

$$\implies \langle w_k, * \rangle = -\frac{1}{2} \cdot \frac{\left[u_k^\top A^{(1)} v_j \right]^2}{\sigma_k} \quad (138)$$

$$\xrightarrow{\text{sum over } j=r+1, \dots, n} - \sum_{j=r+1}^n \frac{1}{2\sigma_k} \left[u_k^\top A^{(1)} v_j \right]^2. \quad (139)$$

Part 6.2.4 (Combine and group terms). Since

$$\sigma_k^{(2,p=1)} = 0, \quad (140)$$

thus,

$$\sigma_k^{(2)} = \sigma_k^{(2,p=2)} \quad (141)$$

$$= - \left[\sum_{i=1}^r \frac{1}{4(\sigma_i - \sigma_k)} \left[\left[u_i^\top A^{(1)} v_k \right]^2 + 2u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \left[u_k^\top A^{(1)} v_i \right]^2 \right] \right. \quad (142)$$

$$\left. + \sum_{i=1}^r \frac{1}{4(-\sigma_i - \sigma_k)} \left[\left[u_i^\top A^{(1)} v_k \right]^2 - 2u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \left[u_k^\top A^{(1)} v_i \right]^2 \right] \right. \quad (143)$$

$$\left. - \sum_{j=r+1}^m \frac{1}{2\sigma_k} \left[u_j^\top A^{(1)} v_k \right]^2 - \sum_{j=r+1}^n \frac{1}{2\sigma_k} \left[u_k^\top A^{(1)} v_j \right]^2 \right] \quad (144)$$

$$= - \left[\sum_{i=1}^r \frac{(\sigma_i + \sigma_k) - (\sigma_i - \sigma_k)}{4(\sigma_i^2 - \sigma_k^2)} \left[\left[u_i^\top A^{(1)} v_k \right]^2 + \left[u_k^\top A^{(1)} v_i \right]^2 \right] \right. \quad (145)$$

$$\left. + \frac{1}{2} \sum_{i=1}^r \frac{(\sigma_i + \sigma_k) + (\sigma_i - \sigma_k)}{\sigma_i^2 - \sigma_k^2} u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i \right. \quad (146)$$

$$\left. - \sum_{j=r+1}^m \frac{1}{2\sigma_k} \left[u_j^\top A^{(1)} v_k \right]^2 - \sum_{j=r+1}^n \frac{1}{2\sigma_k} \left[u_k^\top A^{(1)} v_j \right]^2 \right] \quad (147)$$

$$= \frac{1}{2} \sum_{i \neq k}^r \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} \left[u_i^\top A^{(1)} v_k \right]^2 + \frac{1}{2} \sum_{i \neq k}^r \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} \left[u_k^\top A^{(1)} v_i \right]^2 \quad (148)$$

$$+ \frac{1}{2} \sum_{i=1}^r \frac{\sigma_i}{\sigma_k^2 - \sigma_i^2} u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \frac{1}{2} \sum_{i=1}^r \frac{\sigma_i}{\sigma_k^2 - \sigma_i^2} u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i \quad (149)$$

$$+ \frac{1}{2} \sum_{j=r+1}^m \frac{\sigma_k}{\sigma_k^2 - \sigma_j^2} \left[u_j^\top A^{(1)} v_k \right]^2 + \frac{1}{2} \sum_{j=r+1}^n \frac{\sigma_k}{\sigma_k^2 - \sigma_j^2} \left[u_k^\top A^{(1)} v_j \right]^2 \quad (150)$$

$$= \frac{1}{2} \sum_{i \neq k}^r \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} \left[u_i^\top A^{(1)} v_k \right]^2 + \frac{1}{2} \sum_{i \neq k}^r \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} \left[u_k^\top A^{(1)} v_i \right]^2 \quad (151)$$

$$+ \frac{1}{2} \sum_{i=1}^r \frac{\sigma_i}{\sigma_i^2 - \sigma_k^2} u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i + \frac{1}{2} \sum_{i=1}^r \frac{\sigma_i}{\sigma_i^2 - \sigma_k^2} u_k^\top A^{(1)} v_i u_i^\top A^{(1)} v_k. \quad (152)$$

Part 6.2.5 (Derive Kronecker-Product Representation for Hessian). Use the identities:

1. $\text{vec}(BV A^\top) = (A \otimes B) \text{vec}(V)$,
2. $\mathbb{R} \ni A^\top X B = (A^\top X B)^\top$,

consider:

$$\lim_{x \rightarrow 0} x^2 \left[u_i^\top A^{(1)} v_k \right]^2 = \lim_{x \rightarrow 0} u_i^\top x A^{(1)} v_k u_i^\top x A^{(1)} v_k \quad (153)$$

$$= \left[u_i^\top dA v_k \right] \left[u_i^\top dA v_k \right] \quad (154)$$

$$= \left[v_k^\top dA u_i \right] \left[u_i^\top dA v_k \right] \quad (155)$$

$$= \text{vec} \left[v_k^\top dA u_i \right] \text{vec} \left[u_i^\top dA v_k \right] \quad (156)$$

$$= (u_i^\top \otimes v_k^\top) \text{vec}(dA) (v_k^\top \otimes u_i^\top) \text{vec}(dA) \quad (157)$$

$$= \left[(u_i^\top \otimes v_k^\top) \text{vec}(dA) \right]^\top \left[(v_k^\top \otimes u_i^\top) \text{vec}(dA) \right] \quad (158)$$

$$= \text{vec}(dA^\top) (v_k \otimes u_i) (v_k \otimes u_i)^\top \text{vec}(dA). \quad (159)$$

Similarly,

$$\lim_{x \rightarrow 0} x^2 \left[u_k^\top A^{(1)} v_i \right]^2 = \text{vec}(dA^\top) (v_i \otimes u_k) (v_i \otimes u_k)^\top \text{vec}(dA), \quad (160)$$

$$\lim_{x \rightarrow 0} x^2 u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i = \lim_{x \rightarrow 0} u_i^\top x A^{(1)} v_k u_k^\top x A^{(1)} v_i \quad (161)$$

$$= \lim_{x \rightarrow 0} u_i^\top dA v_k u_k^\top dA v_i \quad (162)$$

$$= \text{vec}(dA^\top) (v_k \otimes u_i) (v_i \otimes u_k)^\top \text{vec}(dA), \quad (163)$$

and

$$\lim_{x \rightarrow 0} x^2 u_k^\top A^{(1)} v_i u_i^\top A^{(1)} v_k = \lim_{x \rightarrow 0} u_k^\top x A^{(1)} v_i u_i^\top x A^{(1)} v_k \quad (164)$$

$$= \lim_{x \rightarrow 0} u_k^\top dA v_i u_i^\top dA v_k \quad (165)$$

$$= \text{vec}(dA^\top) (v_i \otimes u_k) (v_k \otimes u_i)^\top \text{vec}(dA). \quad (166)$$

Hence,

$$D^2[dA, dA] = \text{vec}(dA)^\top \frac{\partial}{\partial \text{vec}(A)} \text{vec} \left(\frac{\partial \sigma_k}{\partial A} \right) \text{vec}(dA) \quad (167)$$

$$= 2 \lim_{x \rightarrow 0} x^2 \sigma_k^{(2)} \quad (168)$$

$$= \sum_{i \neq k} \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} \lim_{x \rightarrow 0} x^2 \left[u_i^\top A^{(1)} v_k \right]^2 \quad (169)$$

$$+ \sum_{i \neq k} \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} \lim_{x \rightarrow 0} x^2 \left[u_k^\top A^{(1)} v_i \right]^2 \quad (170)$$

$$+ \sum_{i=1}^r \frac{\sigma_i}{\sigma_i^2 - \sigma_k^2} \lim_{x \rightarrow 0} x^2 u_i^\top A^{(1)} v_k u_k^\top A^{(1)} v_i \quad (171)$$

$$+ \sum_{i=1}^r \frac{\sigma_i}{\sigma_i^2 - \sigma_k^2} \lim_{x \rightarrow 0} x^2 u_k^\top A^{(1)} v_i u_i^\top A^{(1)} v_k \quad (172)$$

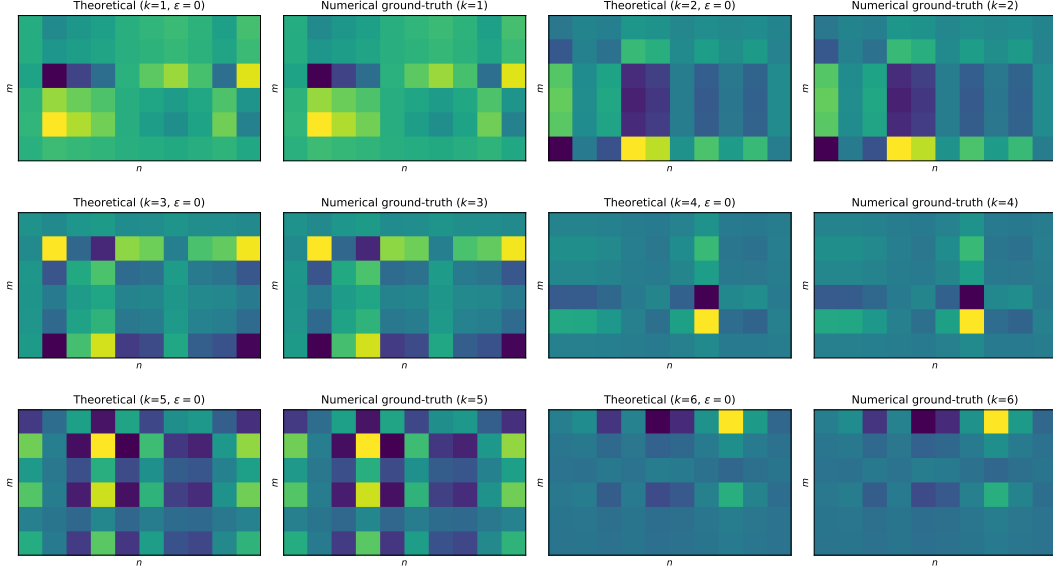
$$\xrightarrow{\text{re-label indices}} \text{vec}(dA^\top) \left[\underbrace{\sum_{i \neq k, i \leq m} \frac{\sigma_k}{\sigma_k^2 - \sigma_i^2} (v_k \otimes u_i) (v_k \otimes u_i)^\top}_{\text{left}} + \right. \quad (173)$$

$$\underbrace{\sum_{j \neq k, j \leq n} \frac{\sigma_k}{\sigma_k^2 - \sigma_j^2} (v_j \otimes u_k) (v_j \otimes u_k)^\top}_{\text{right}} + \quad (174)$$

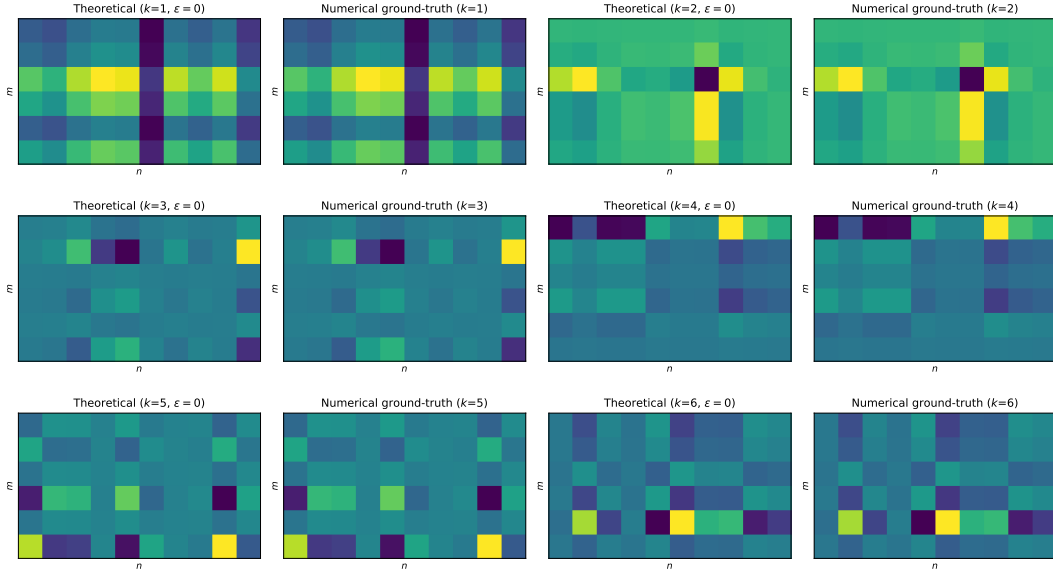
$$\left. \underbrace{\sum_{l \neq k, l \leq r} \frac{\sigma_l}{\sigma_k^2 - \sigma_l^2} \left[(v_k \otimes u_l) (v_l \otimes u_k)^\top + (v_l \otimes u_k) (v_k \otimes u_l)^\top \right]}_{\text{left-right interaction}} \right] \text{vec}(dA) \quad (175)$$

7 Numerical Experiments

We conduct numerical experiments for verifying the correctness of the special cases with $n = 1$ and $n = 2$. We sample matrices from two distributions $\mathcal{N}(0, 1)$ and $U(0, 1)$, respectively. We set random seed to 1 and choose matrix dimensions by 6×10 . The results of Jacobian of singular values are shown in Figure 2; the results of Hessian of singular values are shown in Figure 3. The numerical ground-truths are computed using pytorch's autograd framework. The theoretical results are computed using the formulas we have derived.



(a) The matrix is sampled from $\mathcal{N}(0, 1)$.

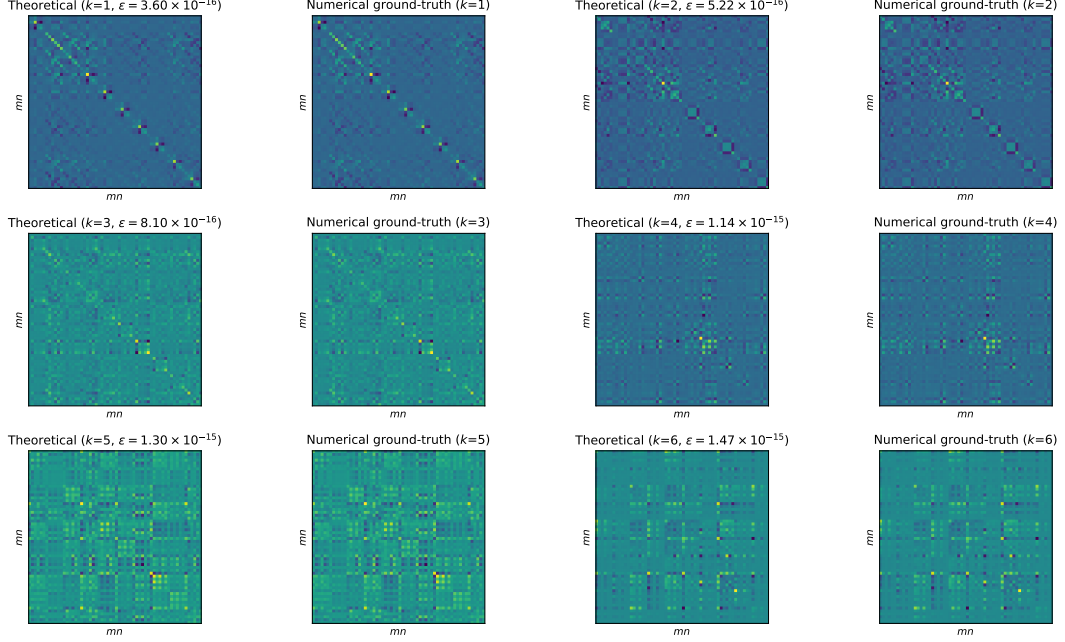


(b) The matrix is sampled from $U[0, 1]$.

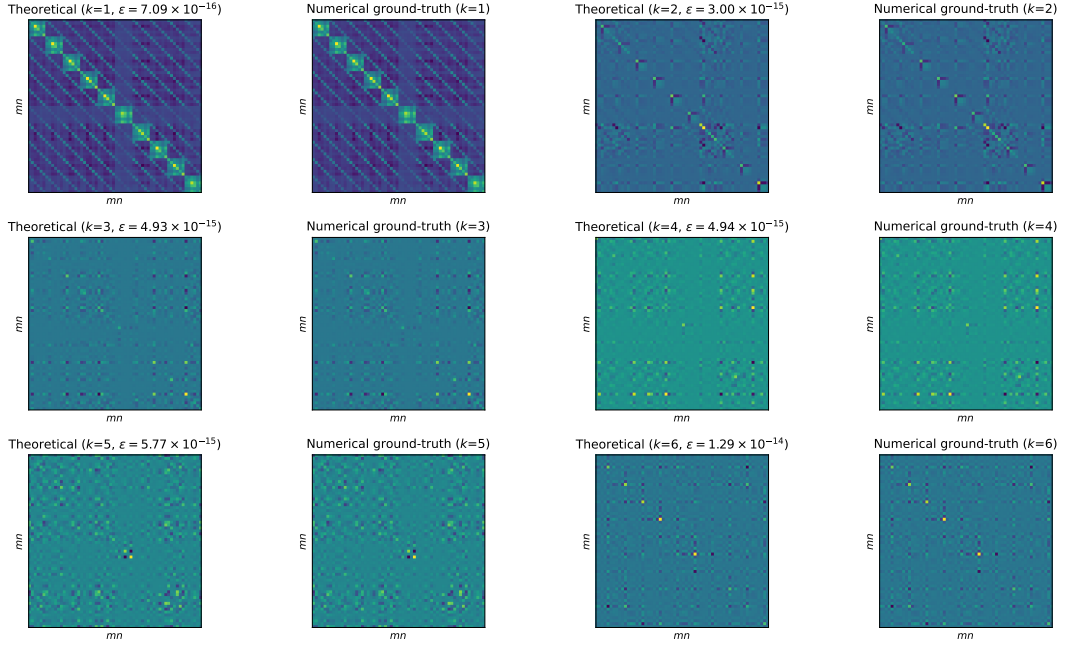
Figure 2: Jacobians of singular values.

8 Conclusion

We present a unified operator-theoretic framework for obtaining closed-form expressions of the infinitesimal n -th order variations of singular values in real rectangular matrices. By viewing matrices as compact operators and extending Kato’s perturbation theory, we derive an explicit formula valid for all n , recover the classical Jacobian when $n = 1$, and obtain a Kronecker-product representation of the Hessian when $n = 2$. Our approach provides practitioners with higher-order sensitivity tools for applications such as adversarial-perturbation analysis in deep learning.



(a) The matrix is sampled from $\mathcal{N}(0, 1)$.



(b) The matrix is sampled from $U[0, 1]$.

Figure 3: Hessians of singular values.

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References

- Alan Edelman and N Raj Rao. Random matrix theory. *Acta numerica*, 14:233–297, 2005.
- Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*, 2014.
- Roger A Horn and Charles R Johnson. *Matrix analysis*. Cambridge university press, 2012.
- Tosio Kato. *Perturbation theory for linear operators*. Springer, 1995.
- Tamara G Kolda and Brett W Bader. Tensor decompositions and applications. *SIAM review*, 51(3): 455–500, 2009.
- Roisin Luo, Alexandru Drimbarean, James McDermott, and Colm O’Riordan. Reclaiming residual knowledge: A novel paradigm to low-bit quantization. In *35th British Machine Vision Conference 2024, BMVC 2024, Glasgow, UK, November 25-28, 2024*. BMVA, 2024a. URL <https://papers.bmvc2024.org/0184.pdf>.
- Roisin Luo, James McDermott, and Colm O’Riordan. Interpreting global perturbation robustness of image models using axiomatic spectral importance decomposition. *Transactions on Machine Learning Research*, 2024b. ISSN 2835-8856. URL <https://openreview.net/forum?id=uQYomAuo7M>.
- Jan R Magnus and Heinz Neudecker. *Matrix differential calculus with applications in statistics and econometrics*. John Wiley & Sons, 2019.
- VA Marchenko and Leonid A Pastur. Distribution of eigenvalues for some sets of random matrices. *Mat. Sb.(NS)*, 72(114):4, 1967.
- Madan Lal Mehta. *Random matrices*, volume 142. Elsevier, 2004.
- Walter Rudin. Principles of mathematical analysis. 2021.
- Michael Spivak. *Calculus on manifolds: a modern approach to classical theorems of advanced calculus*. CRC press, 2018.
- Gilbert W Stewart and Ji-guang Sun. Matrix perturbation theory. (*No Title*), 1990.
- Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. Intriguing properties of neural networks, 2014. URL <https://arxiv.org/abs/1312.6199>.
- Craig A Tracy and Harold Widom. Level-spacing distributions and the airy kernel. *Communications in Mathematical Physics*, 159:151–174, 1994.
- Per-Åke Wedin. Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics*, 12:99–111, 1972.
- Dong Yin, Ramchandran Kannan, and Peter Bartlett. Rademacher complexity for adversarially robust generalization. In *International conference on machine learning*, pages 7085–7094. PMLR, 2019.